Synopsis

Title	Rituximab Therapy for the Induction of Remission and Tolerance in ANCA-Associated Vasculitis
Short Title	*Rituximab for ANCA-Associated Vasculitis*
Clinical Phase	II/III
IND Sponsor	DAIT, NIAID
Study Conduct	ITN
Protocol Chairs	Ulrich Specks, MD John H. Stone, MD
Accrual Objective	200 participants
Accrual Period	30 months
Study Duration	The common closing date will be 18 months after the last participant is enrolled in the trial.
Study Design	This is a randomized, multicenter, double-masked, placebo-controlled trial in participants with severe ANCA-associated vasculitis (AAV). Two hundred participants will be randomized in a 1:1 ratio to the experimental arm or the control arm of the trial.
Primary Study Objective	To determine the efficacy of rituximab (375 mg/m^2, four weekly infusions) and glucocorticoids in the induction of complete remission, defined as a Birmingham Vasculitis Activity Score for Wegener's Granulomatosis (BVAS/WG) of 0 off glucocorticoid therapy.
Primary Endpoint	The percentage of participants who have a BVAS/WG of 0 and have successfully completed the glucocorticoid taper at 6 months after randomization.
Secondary Objectives	1. To compare the safety profile of rituximab with that of conventional therapy. 2. To assess the superiority of rituximab as compared to conventional therapy. 3. To determine the duration of complete remission induced by four infusions of rituximab (375 mg/m2). 4. To determine if participants with severe AAV who are treated with rituximab can achieve clinical tolerance as defined in section 9.
Secondary Endpoints	1. The adverse event rate, expressed as adverse events per participant-year during the 6, 12, and 18 months after randomization for the following adverse events combined: • Death (all causes) • Grade 2 or higher leukopenia or thrombocytopenia • Grade 3 or higher infections

- Hemorrhagic cystitis (grade 2 or lower needs confirmation by cystoscopy)
- Malignancy
- Venous thromboembolic event (deep venous thrombosis or pulmonary embolism).
- Hospitalization resulting either from the disease or from a complication due to study treatment.
- Infusion reactions (within 24 hours of infusion) that result in the cessation of further infusions (including cytokine-release allergic reaction).
- Cerebrovascular accident (CVA).

2. The two-sided 95% CI of the percentage of participants who have a BVAS/WG of 0 and have successfully completed the glucocorticoid taper by 6 months after randomization and two-sided 95% CI of the difference between the two arms.
3. The duration of complete remission, the time to limited and/or severe flare after complete remission (see section 3.7 for the definitions of limited and severe flare) in the two treatment groups.
4. The percentage of participants who meet the criteria for clinical tolerance, as defined in section 1.6.4, at 12 and 18 months after randomization and at the common closing date.

Tertiary Objectives

1. To determine the percentage of participants in complete remission at 12 and 18 months after randomization.
2. To assess other measures of efficacy and safety of rituximab in participants with severe AAV.
3. To determine the effect of rituximab on markers of inflammation and specific immune parameters through a series of detailed mechanistic studies (section 9).

Tertiary Endpoints

1. The percentage of participants in complete remission at 12 and 18 months after randomization.
2. The cumulative BVAS/WG AUC during the 6, 12, and 18 months after randomization.
3. The percentage of participants who achieve and maintain partial remission (defined as having a BVAS/WG of 1 or 2) at months 6, 12, and 18.
4. The percentage of participants who achieve a BVAS of 0 on prednisone <10 mg/day at 6, 12, and 18 months after randomization.
5. The percentage of participants who achieve complete remission after blinded crossover.
6. The cumulative steroid dose between groups for participants at 6, 12, and 18 months.

7. The number of severe flares in participants at 6, 12, and 18 months.
8. The number of limited flares in participants at 6, 12, and 18 months.
9. The percentage of participants who withdraw from the study or treatment because of drug intolerance (e.g., emesis, infusion reactions).
10. Laboratory markers of inflammation (ESR and CRP).

Inclusion Criteria

Potential participants must meet all of the following criteria to be eligible for enrollment in the study:

1. Age: They must be 15 years of age or older.
2. Weight: They must weigh at least 40 kg.
3. Diagnosis type: They must be diagnosed with WG or MPA according to the definitions of the Chapel Hill Consensus Conference (Appendix 7). The maximum number of participants with MPA diagnosis shall not exceed 50% of all participants.
4. Screening diagnosis: They must be newly diagnosed patients, or they must have a disease flare that fulfills inclusion criteria 5, 6, and 7.
5. Disease activity: They must have active disease with a BVAS/WG ≥ 3 that would normally require treatment with CYC.
6. Disease severity: They must have severe disease, i.e., one or more of the major BVAS/WG items listed in Table 2, or disease severe enough to require treatment with CYC.
7. ANCA status: They must be positive for either PR3-ANCA or MPO-ANCA at the screening
8. Contraception contract: They must be willing to practice medically acceptable contraception (e.g., combination barrier method and spermicide, hormonal therapy) until one full menstrual cycle (women) or 3 months (men) have passed after the discontinuation of AZA/AZA placebo and at least 1 year after the first rituximab/rituximab placebo infusion.
9. Pregnancy reporting contract: They must be willing to report pregnancies (females and male's partners) that occur at any time during the trial and up to 1 year after completion of study therapy.
10. Breastfeeding contract: If female, they must be willing to refrain from breastfeeding throughout the trial.
11. Compliance contract: They must be willing to comply with study procedures, including completion of all baseline assessments and mechanistic studies within 14 days of starting intravenous steroids for a disease flare, with the assistance of a caregiver if necessary.
12. Competence: They must be willing and able to provide informed consent.

Exclusion Criteria

Patients who meet any of these criteria will not be enrolled in this study:

1. Diagnosis: They are diagnosed with Churg Strauss syndrome as defined by the Chapel Hill Consensus Conference (Appendix 7).
2. Disease severity:
    a. Limited disease: They have limited disease that would not normally be treated with CYC.
    b. Severe disease: They require mechanical ventilation because of alveolar hemorrhage.
3. Co-morbidities:
    a. Allergies: They have a history of severe allergic reactions to human or chimeric monoclonal antibodies or murine protein.
    b. Infection (systemic): They have an active systemic infection.
    c. Infection (deep space): They have, or have been diagnosed as having, a deep-space infection, such as osteomyelitis, septic arthritis, or pneumonia complicated by empyema or lung abscesses, within 6 months of randomization.
    d. Infection (blood borne): They have active hepatitis B or active hepatitis C or a documented history of HIV, hepatitis B, or hepatitis C.
    e. Liver disease: They have acute or chronic liver disease that is deemed sufficiently severe to impair their ability to participate in the trial.
    f. Renal disease: They have a history of documented anti-GBM disease.
    g. Malignancy: Active or history of malignancy in the last 5 years. Individuals with squamous cell or basal cell skin carcinomas and individuals with cervical carcinoma in situ may be enrolled if they have received curative surgical treatment.
    h. Uncontrolled disease: They show evidence of other uncontrolled disease, including drug and alcohol abuse, that could interfere with participation in the trial according to the protocol.
4. Diagnostics:
    a. WBCs: They have a white blood cell count that is less than 4,000/mm^3.
    b. Platelets: They have a platelet count that is less than 120,000/mm^3.
    c. Liver function tests: They have an ALT or AST level

greater than 2.5 times the upper limit of normal that cannot be attributed to underlying AAV disease.

d. Creatinine: They have a serum creatinine level greater than 4.0 mg/dL that is attributed to renal failure from a current flare. Individuals with stable renal failure from the previous episode of active disease may be included in the study if the flare involves other organ systems.

e. Human antichimeric antibodies: They have had any history of HACA formation.

f. Pregnancy test: positive

5. Treatments

a. CYC (adverse effects): They are intolerant to CYC—i.e., they have previously suffered the adverse effects of conventional therapy that preclude CYC use. These effects may include CYC-induced hemorrhagic cystitis, bone marrow hypoplasia, or malignancy.

b. CYC (recent use): They have used CYC, oral or IV, within the past 4 months unless they started oral CYC not more than 1 week prior to enrollment (see section 5.6.2).

c. Monoclonal antibodies: They have had any previous treatment with rituximab or Campath-1H.

d. Prohibited medications: They have used any of the prohibited medication listed in section 5.6.2.

e. Plasma exchange: They have been treated with plasma exchange within the 3 months preceding the screening visit.

f. Vaccines: They have had a live vaccine fewer than 4 weeks before randomization.

Summary of Study Procedures

Screening: procedures to establish inclusion/exclusion criteria.

Baseline: procedures to establish baseline values for efficacy outcome (BVAS/WG) and other outcomes of interest: immediately after randomization but before receiving any treatment.

Remission induction phase: from the first administration of study treatment through month 6. BVAS/WG scores, use of prednisone, and the occurrence of selected adverse events obtained during this period will be analyzed for the primary and secondary analyses. Participants may be crossed over to the other treatment arm should they be considered treatment failures (see section 3.8.1.1).

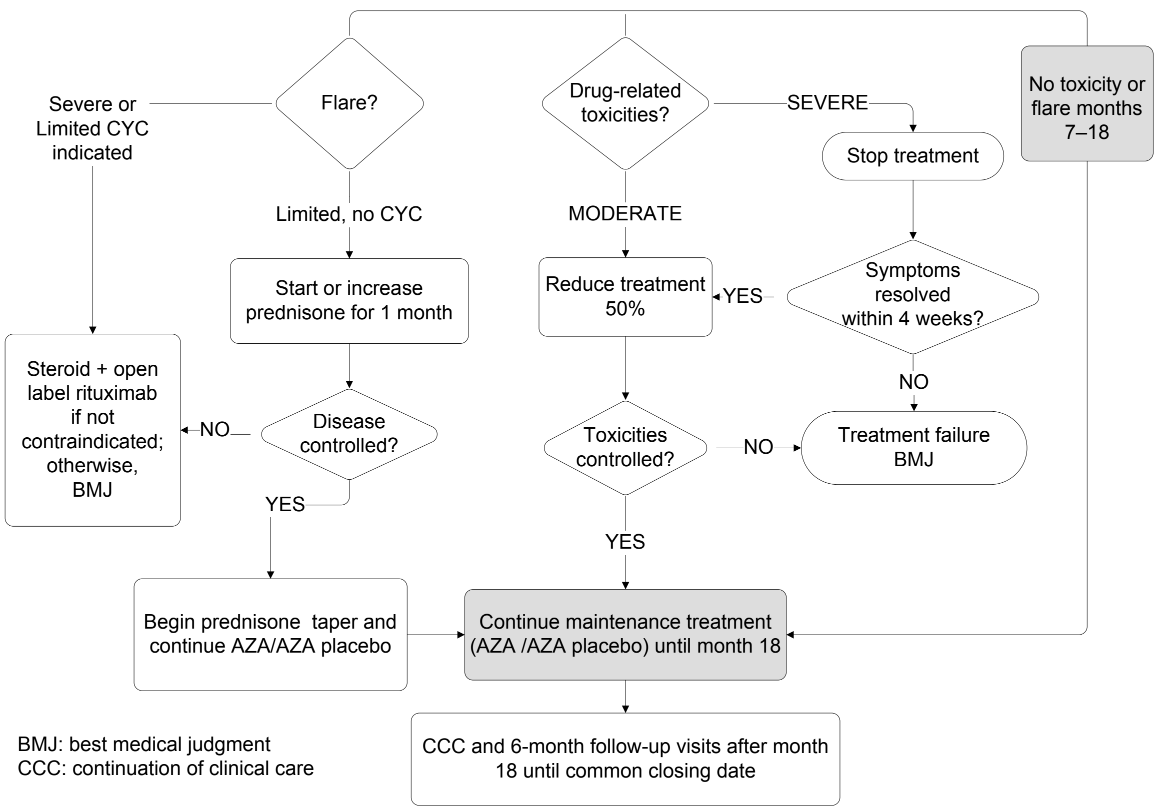
Remission maintenance phase: beginning after month 6 through the common closing date (18 months after enrollment of the last participant). Efficacy and safety outcomes obtained during this period will be analyzed for secondary endpoints including induction of tolerance. Participants who experience severe flares during this period will be treated with open label rituximab (see section 3.8.1.2). After month 18, the investigator may treat participant based on best medical judgment.

Treatment Description	**Glucocorticoids.** Glucocorticoids will be given to both treatment arms. Participants will receive a 1-day course of intravenous glucocorticoids, followed by oral prednisone; the IV glucocorticoids may be repeated up to a maximum of 3 days at the discretion of the investigator. The prednisone will be tapered so that by month 6 all participants in clinical remission will be off glucocorticoids. The guidelines for intravenous glucocorticoid therapy and the prednisone tapering are provided in section 5.3.1. **Remission induction phase (baseline i.e. date of randomization through months 3 to 6):** The experimental arm will receive intravenous infusions of rituximab (375 mg/m2/week times 4) and daily CYC placebo plus oral prednisone. The control arm will receive CYC (2 mg/kg, with doses modified for renal dysfunction) and four weekly infusions of rituximab placebo plus oral prednisone. **Remission maintenance phase (months 3 to 6 through month 18):** The experimental arm will switch from daily CYC placebo to AZA placebo. The control arm will switch from daily CYC to AZA (2 mg/kg/day). After completion of the month 18 visit, treatment will be according to best medical judgment.
Statistical Consideration and Sample Size	**Sample size assumptions**: A 70% complete remission rate in the control group and the experimental group at 6 months after randomization; a noninferiority limit of 20% on the difference in the complete remission percentage between the experimental arm and the control arm; a one-sided 0.025 level test; a 10% dropout rate. These assumptions require 100 participants in each arm to have 83% power to conclude noninferiority. **Primary efficacy analysis:** The difference in the percentage of participants who attain complete remission in the experimental group versus the percentage of participants who attain complete remission in the control group, which will be obtained using a t-distribution multiplier with a one-sided 97.5% confidence interval. The lower bound of this confidence interval around the difference in proportions will be used to assess noninferiority and superiority. If the lower bound is below −20%, noninferiority will be rejected. If the lower bound is above −20%, noninferiority will be concluded. If the point estimate for the complete remission rate in the experimental arm is lower than that of the control arm, the point estimate for the control arm complete remission rate must be at least 40% in order to meet the claim of noninferiority for rituximab. If the lower bound of the difference is above zero, superiority will be concluded on the condition that the lower bound of the two-sided 95% confidence interval of the complete remission rate at 6 months in the experimental arm is greater than or equal to 50%. **Safety analysis:** The rate of the selected adverse events (see secondary endpoints above). A Poisson regression model will be used to compare

the difference of the above adverse events between the two treatment groups.

Other analyses: Kaplan-Meier and Cox proportional hazards model will be used to evaluate differences in the time-to-event endpoints between the two groups. Linear regression models will be used to evaluate changes in continuous outcomes. Repeated measure analyses will be used to account for the within-person correlation.

Months 7–18 AZA/AZA placebo



BMJ: best medical judgment
CCC: continuation of clinical care

Appendix 1. Schedule of Assessments for Original Treatment Assignment

			Week[1]			Month								Post-Treatment Follow-up	
Time Point	**Screen-ing**	**Base-line[2]**	**1**	**2**	**3**	**1**	**2**	**4**	**6**	**9**	**12**	**15**	**18**	**Every 6 months after V12[3]**	**Common closing date**
Visit	**V–1[4]**	**V1**	**V2**	**V3**	**V4**	**V5**	**V6**	**V7**	**V8**	**V9**	**V10**	**V11**	**V12**	**$V_{6\text{-month interval}}$**	**V_{CCD}**
Day	**0**	**1**	**8**	**15**	**22**	**29**	**60**	**120**	**180**	**270**	**365**	**455**	**545**		
General Assessments															
Informed consent	X														
Inclusion and exclusion criteria	X	X													
Randomization		X													
SF-36 v.2™ Health Survey[5,6]	X	X				X	X	X	X	X	X	X	X	X	X
Demography	X														
Medical history (with vaccine)	X														
Skin tests (PPD, anergy panel)[7]	X														
Glucocorticoid log[8]	X	X	X	X	X	X	X	X	X	X	X	X	X		
Concomitant medications[9]	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Adverse events		X	X	X	X	X	X	X	X	X	X	X	X	X	X
Safety officer adverse events[10]		X	X	X	X	X	X	X	X	X	X	X	X		
Vital signs[11]	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Height	X														
Weight	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

[1] For visit windows please refer to section 6.5.
[2] Assessments done within 14 days of baseline visit do not have to be repeated at the discretion of the investigator.
[3] Please record these visits on the Follow-up Visit CRFs and mark these visits sequentially as V13, V14, V15, etc. until V_{ccd}.
[4] The screening visit must be completed within 14 days of the baseline visit.
[5] Should be completed before the participant sees the physician, undergoes any tests or treatment, or receives any tests that day EXCEPTION: Screening Visit.
[6] Ideally should be performed at screening visit. May be performed at either screening visit or baseline visit but not both.
[7] Candida and tetanus booster formulation used. Note: skin test placement is required before starting the baseline infusion. Skin test readings are not required prior to infusion.) To be repeated for testing immunotolerance in a subset of participants after month 6 (see section 9)
[8] All glucocorticoids given as specified by the protocol for treatment of vasculitis should be recorded on the glucocorticoid logs. Glucocorticoids given for any other reason—i.e., for management of asthma or for management of a participant according to BMJ or CCC—should be recorded on the concomitant medications sheet.
[9] Within the last 30 days at the screening visit
[10] Not applicable after V12 or termination visit.
[11] During the infusion, every 15 minutes for 1 hour; then every 30 minutes; and then at least 1 hour after the completion of the infusion.

			Week[1]			Month								Post-Treatment Follow-up	
Time Point	**Screen-ing**	**Base-line[2]**	**1**	**2**	**3**	**1**	**2**	**4**	**6**	**9**	**12**	**15**	**18**	**Every 6 months after V12[3]**	**Common closing date**
Visit	**V–1[4]**	**V1**	**V2**	**V3**	**V4**	**V5**	**V6**	**V7**	**V8**	**V9**	**V10**	**V11**	**V12**	**$V_{6\text{-month interval}}$**	**V_{CCD}**
Day	**0**	**1**	**8**	**15**	**22**	**29**	**60**	**120**	**180**	**270**	**365**	**455**	**545**		
Safety officer assessment[12]		X	X	X	X	X	X	X	X	X	X	X	X		
Physical examination[13]	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Chest x-ray or CT scan[14]	X					X		X	X	X	X	X	X		X
ECG[15]	X										X		X		X
BVAS/WG and flare history[16]	X					X	X	X	X	X	X	X	X	X	X
Physician Global Assessment Form[6]	X	X				X	X	X	X	X	X	X	X	X	X
Treatment questionnaire (MD)									X						
VDI		X							X		X		X	X	X
AVID		X							X		X		X	X	X
Hematology[17]	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Chemistry[18]	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
ANCA (clinical)[19]	X														
TPMT[20]	X														
Serum pregnancy test[21]	X														
UA with microscopy[22]	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

[12] Not applicable after V12 or termination visit.
[13] At baseline/V1 to Week 3/V4 if clinically indicated.
[14] CXR is required at either baseline or screening and months 1, 6, 9, 18 for all participants; and at months 2,4,12 and 15 for participants with abnormal CXR at any visit. CXRs obtained within 2 weeks of a study visit does not have to be repeated at the investigator's discretion. CT scans can be performed instead of CXRs at the investigator's discretion.

[15] After screening, an ECG must be done at V10, V12, and when the participant withdraws or terminates from the study.
[16] Flare history not required at screening and baseline.
[17] STAT on V2-V4. If a pre-infusion WBC <3,000/mm^3 is noticed, the infusion should be withheld; hematology includes WESR; does not need to be repeated at baseline if not clinically indicated
[18] Chemistry includes only BUN, creatinine, and C-reactive protein; it does not need to be repeated at baseline if not clinically indicated
[19] The clinical ANCA testing performed at the screening visit will be done locally and will determine the participant's eligibility.
[20] TPMT specimens will be drawn at screening, but results are not required for randomization. TPMT may determined by TPMT testing or a completed AZA course of at least 125 mg/day.
[21] Only for women of child-bearing potential
[22] Does not need to be repeated at baseline if not clinically indicated

			Week[1]			Month								Post-Treatment Follow-up	
Time Point	**Screen-ing**	**Base-line[2]**	**1**	**2**	**3**	**1**	**2**	**4**	**6**	**9**	**12**	**15**	**18**	**Every 6 months after V12[3]**	**Common closing date**
Visit	**V–1[4]**	**V1**	**V2**	**V3**	**V4**	**V5**	**V6**	**V7**	**V8**	**V9**	**V10**	**V11**	**V12**	$V_{\text{6-month interval}}$	V_{CCD}
Day	**0**	**1**	**8**	**15**	**22**	**29**	**60**	**120**	**180**	**270**	**365**	**455**	**545**		
Mechanistic Assays															
PBMC T-cell assay [23]	X					X	X	X	X		X		X	X	X
Whole-blood DNA HLA genotyping							X								
Whole-blood flow cytometry–panel staining [23]	X			X		X	X	X	X	X	X	X	X	X	X
Whole-blood gene expression profiling [23]	X			X		X	X	X	X	X	X		X	X	X
Serum-secreted cytokines [23]	X			X		X	X	X	X	X	X	X	X	X	X
Serum archive [23]	X			X		X	X	X	X	X	X	X	X	X	X
Plasma archive [23]	X			X		X	X	X	X	X	X	X	X	X	X
Serum HACA [23]	X							X	X	X			X	X	X
Serum ANCA [23,24]	X					X	X	X	X	X	X	X	X	X	X
Serum PK (rituximab levels)[25]	X			X		X	X	X	X	X			X	X	X
Medications															
Glucocorticoid IV[26]		X													
Glucocorticoid PO	X	X	X	X	X	X	X	X							
Rituximab/rituximab placebo[27]		X	X	X	X										
Oral study drug kits[28]		X	X	X	X	X	X	X	X	X	X	X			
Prophylactic medications	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

[23] Also to be done at time of flare and switchover (discontinuation of CYC and start of AZA).
[24] For mechanistic assays. Not carried out locally
[25] See the manual of operation.
[26] A maximum of three 1-day glucocorticoid IV doses can be administered. Last glucocorticoid infusion must be given within 14 days before the first rituximab/placebo infusion. Infusion 1 and IV steroid can be administered on the same day. See section 6.1 for the roles of the investigator versus the role of the safety officer during the infusion.
[27] Participants will be premedicated with diphenhydramine (50 mg) and acetaminophen (650 mg) orally 1 hour (plus or minus 15 minutes) before each infusion. The infusion will be administered in a monitored setting, with access to resuscitative drugs, monitoring devices, and CPR equipment.
[28] CYC/CYC placebo during remission induction phase; AZA/AZA placebo during remission maintenance phase. Participants will be switched over from CYC/CYC placebo to AZA/AZA placebo between 3-6 months as per section 6.3.3.1.

Appendix 2. Schedule of Assessments for Crossover Participants

Participants can be crossed over to the alternate drug treatment any time between visit V5 (1 week after the last rituximab/rituximab placebo infusion) and visit V8 (month-6 study visit).Crossover participants will start over at baseline (V1A) and be followed according to the schedule below. All target visit dates for V9 to the common closing date for crossover participants will have the same visit window, but not the same dates that were calculated from V1. All remaining visits will be derived from the V1A visit. In order to get data as close to the time of flare as possible, samples for mechanistic studies should be collected at the visit when the participant is crossed over (this acts as V1A). Clinical samples are to be drawn before infusion either at time of flare or on the day of infusion. Mechanistic samples are not re-collected on day of infusion. Please also refer to the footnotes in Appendix 1.

		Week			Month								Post-treatment Follow-up	
Time Point	**Base-line**	**1**	**2**	**3**	**1**	**2**	**4**	**6**	**9**	**12**	**15**	**18**	**Every 6 months after V12**	**Common closing date**
Visit	**V1A**[1]	**V2A**	**V3A**	**V4A**	**V5A**	**V6A**	**V7A**	**V8A**	**V9**	**V10**	**V11**	**V12**	$V_{6\text{-month interval}}$ [2]	V_{CCD}
Day	**1**	**8**	**15**	**22**	**29**	**60**	**120**	**180**	**270**	**365**	**455**	**545**		
General Assessments														
SF-36 v.2™ Health Survey	X				X	X	X	X	X	X	X	X	X	X
Glucocorticoid log[3]	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Concomitant medications	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Adverse events	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Safety officer adverse events	X	X	X	X	X	X	X	X	X	X	X	X		
Vital signs	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Height	X													
Weight	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Safety officer assessment	X	X	X	X	X	X	X	X	X	X	X	X		
Physical examination	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Chest x-ray or CT scan[4]	X				X		X	X	X	X	X	X		X
ECG	X									X		X		X
BVAS/WG and flare history	X				X	X	X	X	X	X	X	X	X	X

[1] The participant will not need to undergo screening procedures or to observe the washout periods for prohibited medications.
[2] Please record these visits on the Follow-up Visit CRFs and mark these visits sequentially as V13, V14, V15, etc. until the V_{ccd}.
[3] All glucocorticoids given as specified by the protocol for treatment of vasculitis should be recorded on the glucocorticoid logs. Glucocorticoids given for any other reason—i.e., for management of asthma or for management of a participant according to BMJ or CCC—should be recorded on the concomitant medications sheet.
[4] Chest imaging results within 2 weeks of the first crossover visit (V1A) will be used for the BVAS/WG.

		Week			Month								Post-treatment Follow-up	
Time Point	**Base-line**	**1**	**2**	**3**	**1**	**2**	**4**	**6**	**9**	**12**	**15**	**18**	**Every 6 months after V12**	**Common closing date**
Visit	**V1A**[1]	**V2A**	**V3A**	**V4A**	**V5A**	**V6A**	**V7A**	**V8A**	**V9**	**V10**	**V11**	**V12**	$V_{6\text{-month interval}}$ [2]	V_{CCD}
Day	**1**	**8**	**15**	**22**	**29**	**60**	**120**	**180**	**270**	**365**	**455**	**545**		
Physician Global Assessment Form	X				X	X	X	X	X	X	X	X	X	X
Treatment questionnaire (MD)								X						
VDI	X							X		X		X	X	X
AVID	X							X		X		X	X	X
Hematology	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Chemistry	X	X	X	X	X	X	X	X	X	X	X	X	X	X
UA with microscopy	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Mechanistic Assays														
PBMC T-cell assay[5]	X				X	X	X	X		X		X	X	X
Whole-blood DNA HLA genotyping[6]						X								
Whole-blood flow cytometry[5]	X		X		X	X	X	X	X	X	X	X	X	X
Whole-blood gene expression profiling[5]	X		X		X	X	X	X	X	X		X	X	X
Serum-secreted cytokines[5]	X		X		X	X	X	X	X	X	X	X	X	X
Serum archive[5]	X		X		X	X	X	X	X	X	X	X	X	X
Plasma archive[5]	X		X		X	X	X	X	X	X	X	X	X	X
Serum HACA[5]	X						X	X	X			X	X	X
Serum ANCA[5]	X				X	X	X	X	X	X	X	X	X	X
Serum PK (Rituximab levels)	X		X		X	X	X	X	X			X	X	X
Medications														
Glucocorticoid IV	X													
Glucocorticoid PO	X	X	X	X	X	X	X							
Rituximab/rituximab placebo	X	X	X	X										
Oral study drug kits	X	X	X	X	X	X	X	X	X	X	X			

[5] Also to be done at time of flare and switchover (discontinuation of CYC and start of AZA).
[6] Only if this collection has not been made because (1) crossover has occurred before V6 during original treatment or (2) the blood draw has been missed for a reason other than nonconsent.

		Week			Month								Post-treatment Follow-up	
Time Point	**Base-line**	**1**	**2**	**3**	**1**	**2**	**4**	**6**	**9**	**12**	**15**	**18**	**Every 6 months after V12**	**Common closing date**
Visit	**V1A**[1]	**V2A**	**V3A**	**V4A**	**V5A**	**V6A**	**V7A**	**V8A**	**V9**	**V10**	**V11**	**V12**	**$V_{6\text{-month interval}}$**[2]	**V_{CCD}**
Day	**1**	**8**	**15**	**22**	**29**	**60**	**120**	**180**	**270**	**365**	**455**	**545**		
Prophylactic medications	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Appendix 3. Schedule of Assessments for Open-label Rituximab

Participants can receive open-label rituximab any time after visit V8 (month-6 study visit) and before visit V12 (month-18 study visit). Crossover participants can receive open-label rituximab any time after visit V8A (month-6 study visit) and before visit V12A (month-18 study visit). Once the decision to use open-label rituximab has been made, the participants will be followed according to the schedule below. All target visit dates for open-label rituximab participants will have the same visit window, but not the same dates that were calculated from V1. All visits after shifting to open-label rituximab will be derived from the V1B visit. In order to get data as close to the time of flare as possible, samples for mechanistic studies should be collected at the visit when the participant is put on open label (this acts as V1B). Clinical samples are drawn before infusion either at time of flare or on the day of infusion (V1B). Mechanistic samples are not re-collected on day of infusion. Please refer to footnotes in Appendix 1 for additional information regarding assessments.

		Week			**Month**					**Post-treatment Follow-up**	
Time point	**Base-line**	**1**	**2**	**3**	**1**	**2**	**4**	**6**	**12**	**Every 6 months after V12**	**Common closing date**
Visit	**V1B**[1]	**V2B**	**V3B**	**V4B**	**V5B**	**V6B**	**V7B**	**V8B**	**V10B**[2]	$V_{\text{6-month interval}}$[3]	V_{CCD}
Day	**1**	**8**	**15**	**22**	**29**	**60**	**120**	**180**	**365**		
General Assessments											
SF-36 v.2™ Health Survey	X				X	X	X	X	X	X	X
Glucocorticoid log[4]	X	X	X	X	X	X	X	X	X		
Concomitant medications	X	X	X	X	X	X	X	X	X	X	X
Adverse events	X	X	X	X	X	X	X	X	X	X	X
Vital signs	X	X	X	X	X	X	X	X	X	X	X
Height	X										
Weight	X	X	X	X	X	X	X	X	X	X	X
Physical examination	X	X	X	X	X	X	X	X	X	X	X
Chest x-ray or CT scan[5]	X				X		X				X
ECG	X										X
BVAS/WG and flare history	X				X	X	X	X	X	X	X
Physician Global Assessment Form	X				X	X	X	X	X	X	X
VDI	X							X	X	X	X
AVID	X							X	X	X	X
Hematology	X	X	X	X	X	X	X	X	X	X	X
Chemistry	X	X	X	X	X	X	X	X	X	X	X
UA with microscopy	X	X	X	X	X	X	X	X	X	X	X

[1] The participant will not need to undergo screening procedures or to observe the washout periods for prohibited medications.

[2] Month 12 after open-label rituximab is a clinical visit consisting of a mechanistic sample collection and the assessment of any adverse events related to the use of rituximab.

[3] Please record these visits on unscheduled visit CRFs and number these visits sequentially as 6 months after V10B, 12 months after V10B, etc. until the V_{CCD}.

[4] All glucocorticoids given as specified by the protocol for treatment of vasculitis should be recorded on the glucocorticoid logs. Glucocorticoids given for any other reason—i.e., for management of asthma or for management of a participant according to BMJ or CCC—should be recorded on the concomitant medications sheet.

[5] Chest imaging results within 2 weeks of the first open-label rituximab visit (V1B) will be used for the BVAS/WG.

		Week			Month					Post-treatment Follow-up	
Time point	**Base-line**	**1**	**2**	**3**	**1**	**2**	**4**	**6**	**12**	**Every 6 months after V12**	**Common closing date**
Visit	**V1B**[1]	**V2B**	**V3B**	**V4B**	**V5B**	**V6B**	**V7B**	**V8B**	**V10B**[2]	**$V_{6\text{-month interval}}$**[3]	**V_{CCD}**
Day	**1**	**8**	**15**	**22**	**29**	**60**	**120**	**180**	**365**		
Mechanistic Assays											
PBMC T-cell assay[6]	X				X			X			
Whole blood flow cytometry[6]	X		X		X			X	X		
Whole blood gene expression profiling[6]	X		X		X			X			
Serum-secreted cytokines[6]	X		X		X			X			
Serum archive[6]	X		X		X			X			
Plasma archive[6]	X		X		X			X			
Serum ANCA[6]	X		X		X			X			
Medications											
Glucocorticoid IV	X										
Glucocorticoid PO	X	X	X	X	X	X	X				
Rituximab[7]	X	X	X	X							
Prophylactic medications	X	X	X	X	X	X	X	X	X	X	X

[6] Also to be done at time of flare.

[7] Since the participant is not taking oral study drug, the safety officer is no longer involved in the management of this participant (see section 6.1).

Appendix 4. Schedule of Assessments for BMJ

	BMJ Follow-up	
Time point	**Every 6 months after**	**Common closing date**
Visit	$V_{6\text{-month interval}}$[42]	V_{CCD}
General Assessments		
Inclusion and exclusion criteria		
SF-36 v.2™ Health Survey	X	X
Concomitant medications	X	X
Adverse events	X	X
Vital signs	X	X
Height		
Weight	X	X
Physical examination	X	X
Chest x-ray or CT scan		
ECG		
BVAS/WG and flare history	X	X
Physician Global Assessment Form	X	X
VDI	X	X
AVID	X	X
Hematology	X	X
Chemistry	X	X
UA with microscopy	X	X
Medications		
Glucocorticoid IV		
Glucocorticoid PO		
Prophylactic medications	X	X
Rituximab (for AAV)		

[42] Please record these visits on unscheduled visit CRFs and number these visits sequentially as 6 months after BMJ, 12 months after BMJ, etc. until the V_{CCD}.